Device Specifications

Hardware Components of the Low-Cost Smart Raven Deterrent System (Heo, Baumann, Margelisch, Giordano & Magno, ETH Zürich)

This reference document compiles the technical specifications of every physical device used in the raven deterrent prototype, with emphasis on the parameters that were functionally relevant to the system's design decisions — audio frequency response for raven-call detection, compute throughput and memory for on-device TinyML inference, and power/voltage/current characteristics for the solar energy-harvesting budget.

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1. Compute — Sony Spresense Main Board (CXD5602)

Role in project: hexa-core MCU running MFCC feature extraction and CNN inference for raven detection.

Parameter	Specification
Manufacturer / Part No.	Sony Semiconductor Solutions — CXD5602 (board: CXD5602PWBMAIN1)
CPU	6 × ARM Cortex-M4F cores (with FPU), FD-SOI process
Max. clock frequency	156 MHz (selectable low-power modes: 156 MHz / 32 MHz / 8 MHz)
Auxiliary cores	Additional Cortex-M0+ (system) and Cortex-M4 (sensor) cores support the 6 app cores
SRAM (main memory)	1.5 MB shared SRAM
Flash memory	8 MB
Audio codec	192 kHz / 24-bit high-resolution audio codec with Class-D full-digital amplifier
Microphone inputs	Up to 4 analog + 8 digital microphone channels at 192 kHz / 24-bit
Positioning	Integrated multi-GNSS receiver (GPS, QZSS, GLONASS, SBAS, Galileo, BeiDou) — unused in this p
I/O	GPIO, SPI, I2C, UART, PWM, I2S
Board size	50.0 mm × 20.6 mm (chip footprint: 6.5 mm × 6.5 mm)
Power management	Companion CXD5247 PMIC handles power-path control & battery management, enabling deep-sleep /
Why it mattered here	Multi-core parallelism (CMSIS-NN with DOALL threading) let the team run FFT/MFCC + CNN inference in re

Price: $65.00 USD (listed "no longer stocked" at this retailer at time of writing) — Adafruit Industries, product #4419. Source: https://www.adafruit.com/product/4419

2. Audio Sensing — Electret Microphone + Preamp (SparkFun BOB-12758)

Role in project: captures raw ambient audio (raven calls, background/farm noise) for MFCC feature extraction.

Parameter	Specification
Manufacturer / Part No.	SparkFun Electronics — BOB-12758 (Electret Mic Breakout, OPA344-based)
Microphone type	Omnidirectional electret condenser microphone
Frequency response	100 Hz – 10 kHz
Onboard preamplifier gain	60× (fixed-gain OPA344 op-amp preamp)

Parameter	Specification
Supply voltage range	2.7 V – 5.5 V
Output type	Analog amplitude-modulated voltage signal (for MCU ADC input)
Form factor	Small fully-assembled breakout board
Why it mattered here	The 100 Hz–10 kHz passband comfortably covers raven vocalizations (typically in the low-kHz range

Price: $5.95 USD (retail, single unit; volume pricing from $5.06 at 100+ units) — SparkFun Electronics, SKU BOB-12758. Source: https://www.sparkfun.com/sparkfun-electret-microphone-breakout.html

3. Actuation — Speaker

The paper does not specify an exact speaker part number. Functionally, it is a small audio output transducer driven by the Spresense's onboard Class-D full-digital amplifier (192 kHz/24-bit audio path, BTL stereo output capability) to play a deterrent/alarm sound once a raven call is classified. Power draw for actuation is embedded in the measured Phase 1 (active) power profile of 144.2 mW.

Price: Not applicable — no specific speaker part number is given in the paper, so no verifiable price can be sourced for this component.

4. Energy Source — ANYSOLAR IXOLAR Solar Module (SM641K10L)

Role in project: primary energy-harvesting source charging the system's battery for maintenance-free operation.

Parameter	Specification
Manufacturer / Part No.	ANYSOLAR Ltd. (IXOLAR™ series) — SM641K10L
Cell type	Monocrystalline silicon solar cell, Gen3
Cell efficiency	~25% (wafer-level, typical for IXOLAR SolarMD series)
Open-circuit voltage (Voc)	6.91 V
Voltage at max. power (Vmpp)	5.58 V
Short-circuit current (Isc)	268 mA
Current at max. power (Impp)	252 mA
Max. power output (Pmax)	1.404 W
Dimensions	111.0 mm × 63.0 mm × 2.1 mm (0.11 m × 0.06 m as used in the paper)
Surface area (paper's calc.)	6.993 × 10⁻³ m²
Operating temperature	−40 °C to +90 °C
Compliance	RoHS3 compliant
Why it mattered here	The panel's 25% efficiency and physical area were fed directly into the energy-harvesting model: with an a

Price: $29.68 USD (single unit) for the equivalent DFRobot-packaged SM641K10L cell (DFRobot FIT0600, same ANYSOLAR SM641K10L die) — DigiKey Electronics. The bare ANYSOLAR part is otherwise quote/RFQ-only through electronics distributors.
Source: https://www.digikey.com/en/products/detail/dfrobot/FIT0600/9608221

5. Charge Controller — SparkFun Sunny Buddy MPPT Solar Charger

Role in project: maximum-power-point-tracks the solar module's output and manages battery charging.

Parameter	Specification
Manufacturer / Part No.	SparkFun Electronics — Sunny Buddy (PRT-12885)
Charging IC	Analog Devices / Linear Technology LT3652 — 2 A power-tracking battery charging circuit
Charging topology	Maximum Power Point Tracking (MPPT) for single-cell Li-Po/Li-ion
Input voltage range	6 V – 20 V (min. 6 V, max. recommended 20 V)
Default max. charge current	450 mA (adjustable; min. battery rating 450 mAh recommended)
MPPT setpoint adjustment	Onboard potentiometer sets the input "holding voltage" for MPPT tracking
Output	Single-cell Li-Po/Li-ion battery charge output (load connected in parallel with battery)
Connectors	Barrel jack + 2-pin JST input/output, unpopulated screw-terminal pads
Why it mattered here	MPPT tracking lets the charger extract the most usable power from the small 1.4 W IXOLAR panel under

Price: $32.95 USD (retail, single unit; volume pricing from $28.01 at 100+ units) — SparkFun Electronics, SKU PRT-12885.
Source: https://www.sparkfun.com/sparkfun-sunny-buddy-mppt-solar-charger.html

6. Energy Storage — Renata ICP303450PA-02 Li-Ion Polymer Battery

Role in project: single-cell rechargeable battery buffering harvested solar energy to power the node continuously (including overnight).

Parameter	Specification
Manufacturer / Part No.	Renata SA (Switzerland) — ICP303450PA-02
Chemistry	Lithium-ion polymer (rechargeable, single cell)
Nominal voltage	3.7 V
Typical capacity	500 mAh (rated ~510 mAh)
Cycle life	>500 cycles at ≥80% of rated capacity (typical of the Renata ICP series)
Protection	Integrated safety circuit (overcharge / deep-discharge protection)
Dimensions	52 mm × 34.5 mm × 3.5 mm (ultra-thin, low-profile pack)
Why it mattered here	At 3.7 V nominal, this cell matches the Spresense board's native battery-power input and the Sunny Budd

Price: Not publicly listed — distributors carry this exact Renata part on an RFQ (request-for-quote) basis rather than a posted retail price. Source (RFQ product page): https://www.acalbfi.com/se/Power-supplies/Off-grid/Batteries/p/Lithium-Ion-Polymer-Battery--3-7V--510mAh--Rechargeable/0000007S3Y — also listed at https://www.renata.com/en-us/products/lithium-polymer-batteries/icp303450pa-02/

7. Enclosure — 3D-Printed Case

A custom 3D-printed enclosure houses the MCU board, microphone, speaker, solar charger and battery, with an aperture for the solar panel and microphone exposure (see Fig. 3 of the paper, "Exterior" / "Hardware Components"). No material or dimensional datasheet is given in the paper; it is a bespoke mechanical part rather than an off-the-shelf component.

Price: Not applicable — custom 3D-printed part with no commercial listing or price.

8. Evaluation Instrument — Nordic Semiconductor Power Profiler Kit II (PPK2)

Role in project: used only during evaluation, to measure the prototype's real power consumption across its operating phases (not part of the deployed system).

Parameter	Specification
Manufacturer / Part No.	Nordic Semiconductor — nRF-PPK2 (Power Profiler Kit II)
Current measurement range	200 nA – 1 A (ampere-meter mode: 500 nA – 1 A)
Measurement resolution	Auto-ranging, 100 nA – 1 mA depending on range (paper reports ±5 µW resolution at 3
Sampling rate	100 kS/s (10× faster than the original PPK)
Operating modes	Ampere Meter (AMP) mode — DUT powered externally; Source Measure Unit (SMU) mode &#8212
Supply/source voltage range	0.8 V – 5.0 V (software configurable), up to 1 A source current
Host interface	USB (5 V, up to 500 mA per cable; two cables needed for 1 A source mode)
Extra feature	8 digital inputs usable as a low-end logic analyzer for code-synchronized power measurement
Why it mattered here	Its low-current auto-ranging resolution (down to ~100 nA/0.2 µA) let the authors accurately separate

Price: $99.95 USD (single unit; max 3 per customer) — Adafruit Industries, product #5048. Source: https://www.adafruit.com/product/5048

9. Price Summary

All prices below are single-unit retail prices in USD as listed by the cited distributor at the time of writing, and will vary by region, stock, and quantity ordered. Click through the links in each section above to verify current pricing.

Parameter	Specification
Sony Spresense Main Board (CXD5602)	$65.00
SparkFun Electret Mic Breakout (BOB-12758)	$5.95
Speaker	N/A (unspecified part)
ANYSOLAR IXOLAR Solar Module (SM641K10L)	~$29.68 (equivalent DFRobot-packaged part)
SparkFun Sunny Buddy MPPT Charger	$32.95
Renata ICP303450PA-02 Battery	RFQ / price on request
3D-Printed Enclosure	N/A (custom part)
Nordic PPK2 (evaluation instrument only)	$99.95
Approximate deployed-system total (excl. speaker, enclosure, battery, and PPK2)	~$133.58 + RFQ (RFQ is evaluation-only)

Note: Specification values for the Spresense CXD5602, BOB-12758 mic breakout, ANYSOLAR SM641K10L, SparkFun Sunny Buddy, Renata ICP303450PA-02, and Nordic PPK2 are drawn from manufacturer product pages / datasheets, since the paper itself only cites select figures (e.g. clock speed, memory, mic frequency range, panel size). Figures not explicitly stated in the paper are sourced externally and may vary slightly by hardware revision; consult the manufacturer datasheet directly before use in a bill of materials.